

Decision Tree

1- Calculate The Entropy for the table.

$$P_{\text{yes}} = \frac{5}{8} \quad P_{\text{No}} = \frac{3}{8}$$

$$E = -\frac{5}{8} \log_2\left(\frac{5}{8}\right) - \frac{3}{8} \log_2\left(\frac{3}{8}\right) = 0.95$$

First Column:

Region

USA $\begin{cases} P_{\text{yes}} = \frac{2}{4} \\ P_{\text{No}} = \frac{2}{4} \end{cases} \quad E_{\text{USA}} = -\frac{2}{4} \log_2\left(\frac{2}{4}\right) - \frac{2}{4} \log_2\left(\frac{2}{4}\right) = 1$

Canada $\begin{cases} P_{\text{yes}} = \frac{3}{4} \\ P_{\text{No}} = \frac{1}{4} \end{cases} \quad E_{\text{Canada}} = -\frac{3}{4} \log_2\left(\frac{3}{4}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right) = 0.8$

$$\therefore E_{\text{Region}} = \frac{4}{8} \times 1 + \frac{4}{8} \times 0.8 = 0.9$$

$$\therefore \text{I.G} = 0.95 - 0.9 = 0.05$$

Total Entropy

Region Entropy

Get the I.G For each Column, then choose the biggest one to be the parent

D.T for Regression

1- Calculate the mean and MSE.

$$\text{mean} = 100$$

$$\text{MSE} = 200$$

$$\text{mean} = \frac{30+35+45+60+70+50+90}{8} = 100$$

$$\text{MSE} = \frac{1}{n} \sum (x - \bar{x})^2$$

$$\frac{1}{8} (30-100)^2 + (35-100)^2 + (45-100)^2 + \dots = 200$$

Region

USA $\begin{cases} \text{Mean} = \frac{30+45+70+90}{4} = 58 \\ \text{MSE} = 120 \end{cases}$

Canada $\begin{cases} \text{Mean} = 60 \rightarrow \frac{35+60+50}{3} \\ \text{MSE} = 130 \end{cases}$

$$\text{MSE}_{\text{Region}} = \frac{4}{8} \times 120 + \frac{3}{8} \times 130 = 170$$

$$\text{Variance} = \text{MSE}_{\text{total}} - \text{MSE}_{\text{Region}} = 200 - 170 = 30$$

I will get Variance for each one then choose the biggest one to be the Parent.